THE PERTH ARTIFACTORY: STUDIO REPORT

Meg Travers
Western Australian Academy of Performing Arts
Edith Cowan University
Perth, Western Australia, Australia

SKoT McDonald
Perth Artifactory
8/16 Guthrie Street, Osborne Park
Western Australia

ABSTRACT

The Perth Artifactory has become known not only as a 'hacker space', but also as a location for the making and performance of electronic and experimental music and instruments.

Keywords: electronic music, makers, hack, modular synthesizer, digital community, noise, studio report, Theremin, BollART, percussion, FXpansion, BFD.

1. HISTORY

The Perth Artifactory came into existence in 2009. It was originally located in a residential area in an inner suburb of Perth, but soon moved to a large warehouse in the industrial zone of Osborne Park where its members and the range of activities has grown.

The Perth Artifactory has always had a number of members with a strong interest in electronic and experimental music, and in the last three years this has become a substantial sub-group within the organisation, with activities, facilities and regular events that centre around this. There are also many people within this group with ties to the Western Australian Academy of Performing Arts, either as past or current undergraduate and postgraduate students.

2. ACTIVITIES

2.1. NoizeMaschin!!

Entering its fifth year, NoizeMaschin!! (The Perth Artifactory 2015) is a monthly experimental music night organised and curated by a team of people from the Artifactory's music sub-group. The format for NoizeMaschin!! is unusual: each night includes 8 performers (these may be soloists or groups), and each act is given 10 minutes in which to perform, ideally segueing into the following performer. This short-set format means that many styles are heard over each evening, and allows for less "new music fatigue" in the audience, many of whom are newcomers to this genre.

All of the events have been recorded and made available for download (with the performers permission) and are being included in the Western Australian New Music Archive (WANMA).

Whilst the Artifactory is well known to many of the arts funding organisations in Western Australia, it does not receive any ongoing funding from them, and NoizeMaschin!! is produced on a zero dollar budget. The PA system and staging used for the event were found or donated by members who also donate their time to construct them and keep them running. Lighting for the events is often provided by one of the members who has constructed a computer controlled laser display, and also via a hybrid MIDI/DMX system where a controller keyboard is used to literally "play" the lights.

Performers understand that there is no capacity to pay them, however this hasn't dented the enthusiasm of people wishing to perform and it has proved to be a useful avenue for many to trial experimental techniques before moving onto bigger shows.

NoizeMaschin!! also offers three month residencies to new performers which gives them free access to the Artifactory and its facilities, and the opportunity to perform at each NoizeMaschin!! concert over that period.

2.2. Modular Synthesizer building

Another of the regular events is the monthly Modular Synthesizer building night. Local synthesis guru and recent PhD graduate in electronic engineering, Dr Andrew Fitch, designs and builds Eurorack format synthesizer modules (nonlinearcircuits) which are sold worldwide. For these nights however, Andrew supplies the modules in kit form (including all of the components plus PCB board and panel) at very low prices, and assists people to build and test synthesizer modules, which they are literally building themselves.

Another of the members, Nathan Thompson, has designed and cuts inexpensive modular synthesizer cases on the laser cutters; these are also gaining popularity in the modular synthesizer community.

Usually 20-30 people attend these nights each month, with a high percentage of musicians and engineers. This has led to great cross-pollination between the two professions, and the growing possibility that Perth has the highest ratio of DIY modular synth owners in Australia.

2.3. The Sonic BollART

In 2015, the Perth International Arts Festival held the first Arthack in Australia – Hack the Festival (Perth International Arts Festival 2015). This event was a week long hack-a-thon in which ideas were discussed, teams formed, and art hacks created.



Figure 1. Dr Andrew Fitch and one of his nonlinearcircuits modular synthesizers.

The Perth Artifactory had several members on various teams in the event, but also fielded an official Artifactory team which created the Sonic BollART. The BollART was premised upon the idea that bollards are common objects in any urban landscape, serving as a barrier to certain activities or modes of transport, but that it also offers opportunities to house infrastructure for other things. Representing an underappreciated, readily available and numerous art-space with high foot-traffic, a human scale, and a fun set of constraints to fire the imagination, the BollART is a vessel within which a range of experiences can be tailored by individual artists, to create an item to convey their ideas.

The team's entry, named the Sonic BollART, was constructed with the Artifactory's laser cutter and contained a theremin and LED light show to inspire people to physically interact with the object. Whilst theremins are well known, most people have not had an opportunity to try playing one, and its hands-free performance interface provides no barrier to those wishing to attempt to play it.

Addressing the dual problems of distribution for unsigned musicians in the 21st century, and Perth's current issues of finding audiences and venues for local music, The Sonic BollART also hosts a wireless, offline file-sharing service for local musicians to place their recordings, and for any member of the public to download these. In a nod to MOMA's successful *Dead Drops* installation, this service provides a kind of Musical Parkour, giving a direct route for people in an area to discover music and musicians that are around them via a wireless data connection that does not rely on any other infrastructure, nor present a risk of over-utilising expensive 3G and 4G data services. Using the tagline "Un-cloud your music and find your local audience", this system also encourages public discourse on the government monitoring and metadata collection from Internet-based systems.

The Sonic BollART was selected as the winner of the Hack the Festival event, and has since been exhibited at the Totally Huge New Music Festival, local rock music events, street festivals and TEDx.



Figure 2. The Sonic BollART

2.4. Audio and music Making

As a hacker-space, the Perth Artifactory has a range of advanced tools available for members to use, including laser cutters, 3D printers, CNC router and mill, lathe, welders, and so on, as well as a large space in which to design and create. Several musical instruments have been created there.

The Arcophone (Make Magazine 2011) is a MIDI controlled 30,000 volt electric plasma arc instrument which creates audibly tuned arcs of plasma from high voltage arc gaps (or Jacob's Ladders). It is operated using a MIDI controller or software, a little trepidation, and a constant body-position-awareness of how close you are to the plasma arcs.

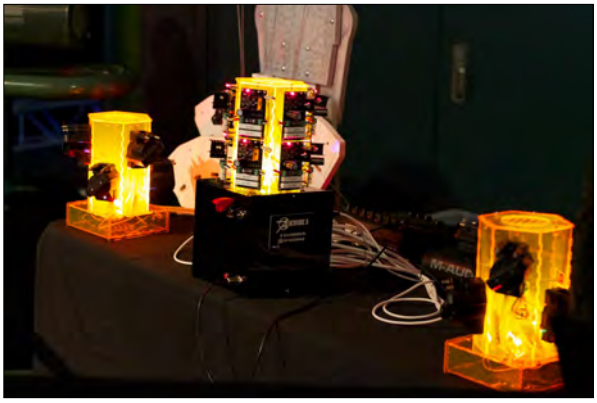


Figure 3. The Arcophone

With many people accessing the space but not necessarily being aware of how to correctly to power up the sound system, the Artifactory also has a network controlled audio system for the stage PA system. This system uses a Raspberry Pi to control power and the timing of switching on equipment via solid state relays. It includes an audio USB card to allow streaming of music from the network (eg: from a user's phone or laptop) and is operated only via a network connection, the hardware deliberately includes no switches or buttons for people to accidentally press or "explore".

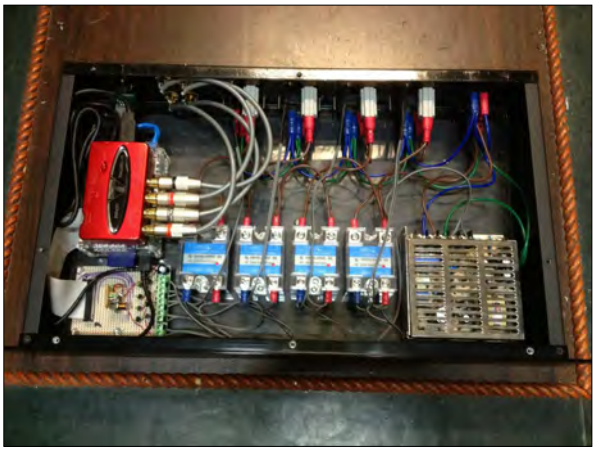


Figure 4. The sound system controller rig.

The Artifactory also hosts a rehearsal room which is operated under a membership system. Up to 5 groups share the rental of that portion of the warehouse giving them a secure, soundproofed room that has a limited number of tenants and the ability to leave large instruments in situ with minimal risk of damage or loss. This has not only assisted in paying the rent on the warehouse, but also provided further networking opportunities for electronic musicians who can promote and organize shared gigs, and to do small scale recordings.

The audio equipment in the room was donated or purchased using part of the proceeds from the Hack the Festival prize money. A local music store has also provided discounts to the users of the room and this has extended the networks of people in both organisations.



Figure 5. The rehearsal room, with obligatory lava lamp and Turkish rug.

Another monthly workshop is Arduino U at which members can be coached or be assisted with Arduino based projects. Following the International Computer Music Conference in Perth in 2013, a number of workshops specifically on Mozzi (Barrass) were held as a result of hearing Tim Barrass speak about this sound synthesis library which he created and maintains for the Arduino.

The construction of physical items is a big part of a hacker space, and with so much music and noise being generated both at the Artifactory and in people's homes, workshops have also been run on building sound dampening baffles. The CNC router *Swarf-o-mat* has also been pressed into use in the making of guitar bodies, and scavenged items have been constructed into a large 'junk percussion' collection which is used by MotET (MotET 2015), one of the bands which make up part of the Artifactory musical community.



Figure 6. Junk percussion rig.

Music Technology has been a strong pillar of activity of the Artifactory since its inception; it was a key interest for several of the founding members; and remains well represented in the governing committee and membership at large to this day. This bringing together of electronic music and maker culture at the Perth Artifactory happens at many levels, from amateur to professional. For example, the current Chairman is also the Chief Technology Officer of music software production company FXpansion (FXpansion); other members have been part of community reach groups with a music focus; instrument makers and so on, with chance and deliberate cross-fertilization between all. It has shown that there are many benefits to both makers and musicians in sharing resources and a diversely skilled community.

3. MUSIC GROUP

Current members of the Perth Artifactory involved in the music group include SKoT McDonald, Sam Gilles, Tim Gilchrist, Andrew Fitch, Nathan Thompson, Meg Travers, Rhys Channing, Josten Myburgh, Steve Paraskos, Stuart McDonald and Jean-Michel Maujean.

4. OTHER MEMBERS

Apart from the music group, members of the Perth Artifactory who have contributed to the projects listed above include:
Wayne Osborne (Lasers for NoizeMaschin!!)
Brett Downing (Construction of Arcophone)
Doug Bateman (Arduino U)
Brendan Ragan (Stage sound system)

5. MORE INFORMATION

http://artifactory.org.au

6. REFERENCES

The Perth Artifactory. *NoizeMaschin!!*. Retrieved 28th September 2015, available online at URL noizemaschin.artifactory.org.au/

nonlinearcircuits. *Analogue Modular Synthesizers*. Retrieved 28th September 2015, available online at URL nonlinearcircuits.com/

Perth International Arts Festival. *Hack the Festival*. Retrieved 28th September 2015, available online at URL 2015.perthfestival.com.au/Whats-on-by-Genre/Theatre-and-Circus/Hack-the-Festival

Make Magazine. *Arcophone Mark II Debuts*. Retrieved 28th September 2015, available online at URL makezine.com/2011/03/14/arcophone_mkii_debuts/

Tim Barrass. *Mozzi*. Retrieved 28th September 2015, available online at URL sensorium.github.io/Mozzi/

MotET. *So come down to the lab…*. Retrieved 28th September 2015, available online at URL www.motet.net.au

FXpansion. *FXpansion*. Retrieved 28th September 2015, available online at URL www.fxpansion.com/